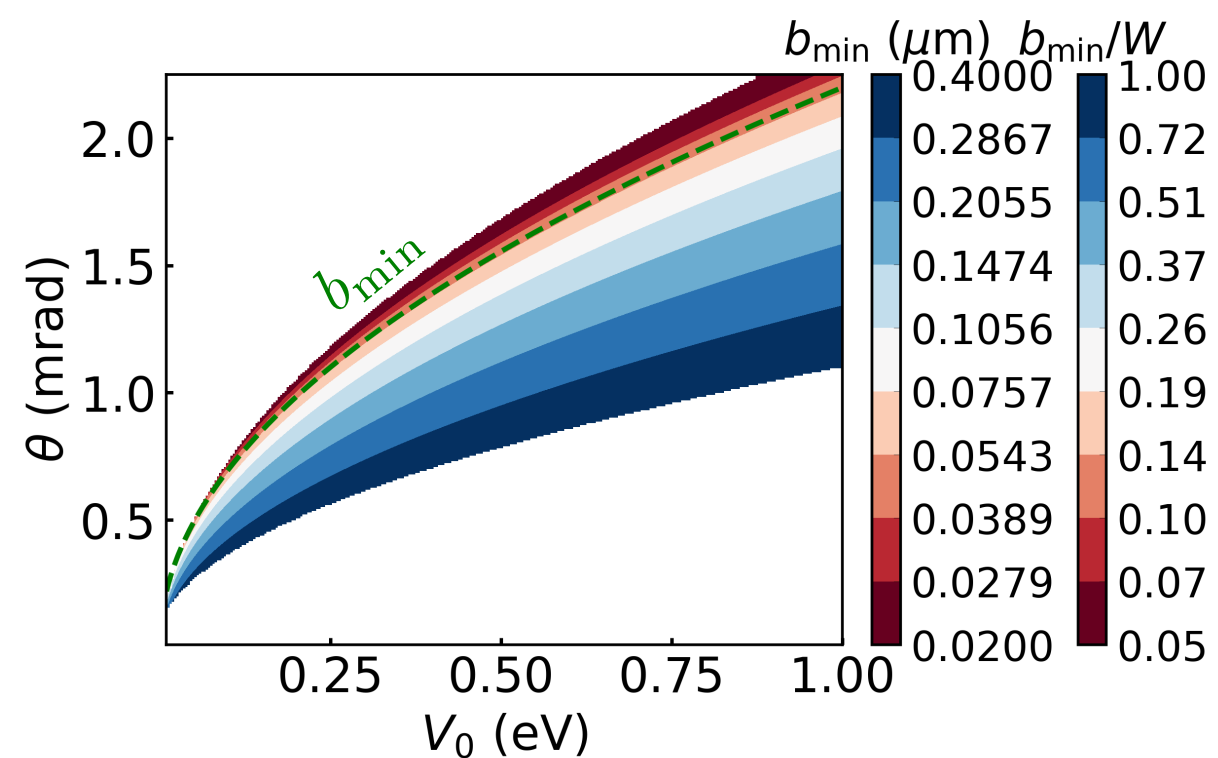
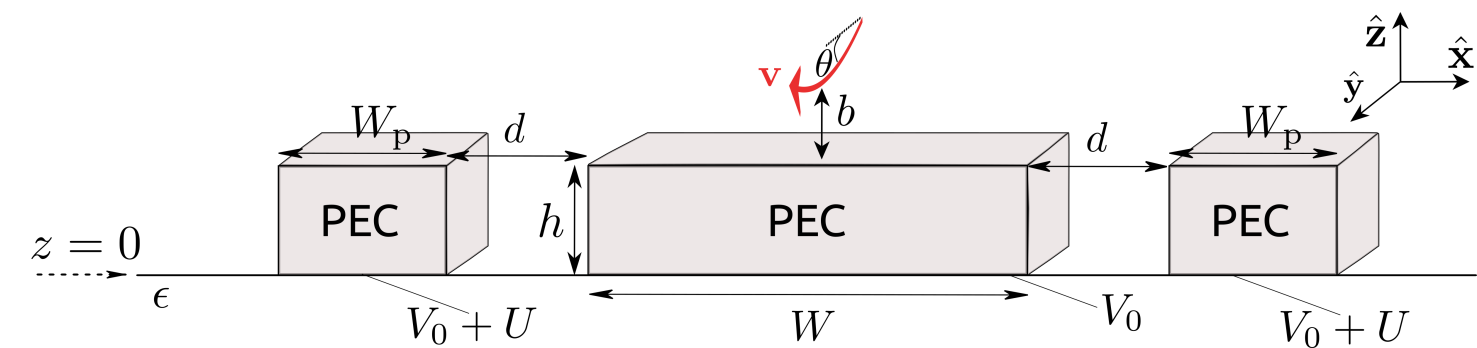


Electron-coupling to waveguide

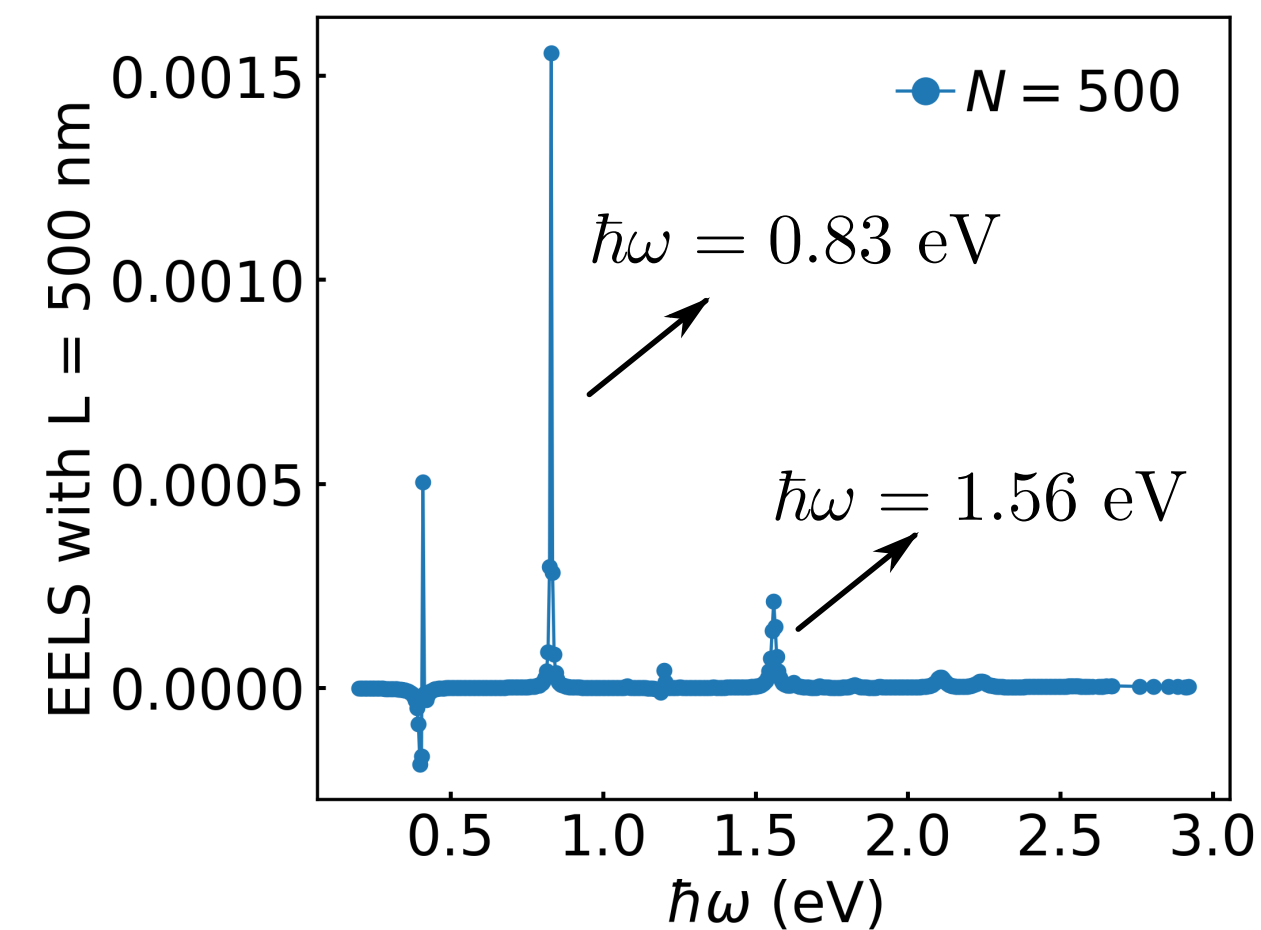


$$E_e = 200 \text{ keV}$$

$$W = 400 \text{ nm } h = 200 \text{ nm}$$

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{b_{\min}+h}^{\infty} \frac{v dz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x, z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy}$$

from bem2d:



$n = 1$

$n = 2$

